Wholesale customer segmentation

Hanu Varma Pinamaraju · EAS509 Project 1 · October 27, 2026

Question and decision

Do annual purchasing patterns support repeatable account groups that would help a wholesale distributor review its customer portfolio? The unit is one customer account. A small set of purchasing profiles could make comparisons easier for account managers. That would be useful only if the profiles are understandable and survive reasonable challenges. A split driven mainly by a few unusual accounts would instead favor individual account review.

Recommendation: this analysis does not yet justify adopting broad customer segments. The selected split isolates 2 accounts and leaves the other 438 undifferentiated.

Data and representation

The data are 440 historical customer accounts in Cardoso's public UCI Wholesale customers CSV [1]. Local labels `uci_001` through `uci_440` preserve source-row order for matching and are excluded from distance. This is the public data version; its correspondence to the course-provided file and `customer_id` values has not been verified.

All 440 rows were eligible and included in every full-data comparison. Eligibility required six finite, nonnegative spending values and a known Channel. There were 0 missing spending values, 0 invalid spending rows, and 0 repeated spending profiles. Channel is nominal: 1 = Horeca and 2 = Retail. No outliers were removed. Spending is in monetary units (m.u.); the currency is unspecified.

The baseline uses **Fresh, Milk, Grocery, Frozen, Detergents_Paper, and Delicassen**. Each is transformed as `log1p`, then centered and divided by its full-sample standard deviation. Accounts are close in Euclidean distance when their standardized log spending amounts are similar. This compresses very large amounts and gives each category equal variance weight. Both purchasing level and category mix remain relevant: accounts are not normalized into spending shares. Correlated categories can still reinforce each other. Channel is reserved for the sensitivity audit; IDs and protected Region are excluded from fitting and selection.

Two concepts of a group

Both methods used the same distance matrix, trying $k = 2, 3, 4, 5$. PAM groups accounts around representative observed accounts, or medoids [2]. Average linkage merges groups using their average between-account distance [3]. Silhouette compares within-group distance with distance to the nearest other group; values near zero indicate weak separation.

Method	Best k	Mean silhouette	Group sizes
PAM	2	0.282	193/247
Average linkage	2	0.569	438/2

Both partitions are scored on the same accounts and distances, so silhouette provides a common descriptive comparison. The highest mean selected average linkage at $k = 2$ for closer examination; exact ties favor smaller k, then PAM. The method, cut, distance, and scaling stayed fixed for the checks below. Both methods assign every account to a group.

What the selected split describes

The two-account group is just 0.45% of the sample. Its low Grocery and Detergents_Paper spending distinguishes it from the remaining accounts, but it is too small and internally uneven to support a broad customer category. The table gives medians in the original monetary units; with two records, a median is their midpoint and need not describe either actual account.

Group	n	Fresh	Milk	Grocery	Frozen	Det/Paper	Deli
Rare pair	2	10,510.0	596.0	70.0	2,241.0	5.0	491.5
Remaining accounts	438	8,504.0	3,641.0	4,785.5	1,526.0	820.5	965.5

Det/Paper abbreviates Detergents_Paper; Deli abbreviates Delicassen. All six spending columns are medians in m.u.

The individual records show why the median needs care. uci_076 spends 20,398 on Fresh, versus 622 for uci_155. Their Grocery values are 3 and 137; Detergents_Paper values are 3 and 7. They share very low spending in selected categories but differ substantially elsewhere. Their transaction records would need checking before treating the pair as a customer type.

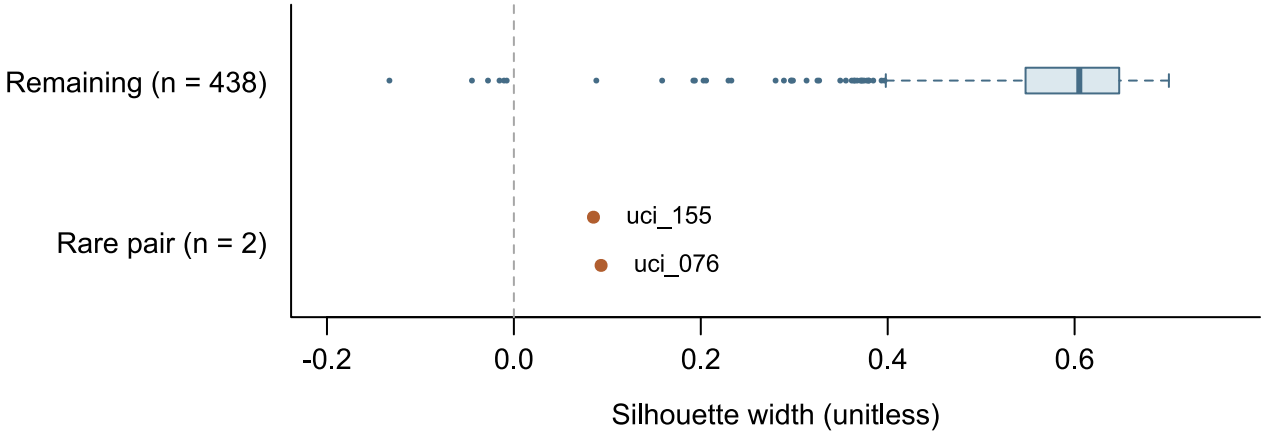


Figure 1: Silhouette for the selected fit. Orange points show the rare accounts separately. The blue box marks the other 438 accounts' median and interquartile range; whiskers reach data within 1.5 IQR, with dots beyond them. Source: public UCI data.

The overall mean silhouette is 0.569, but the rare pair's mean is only 0.089, versus 0.571 for the other 438 accounts. There are 6 negative values overall. The large group dominates the average. Its favorable score is relative to the rare pair; it does not show that all 438 accounts share one meaningful purchasing profile.

PAM at $k = 2$ gives a 193/247 split with mean silhouette 0.282. Agreement with the selected cut is approximately chance level (ARI −0.002). ARI compares pairs of accounts, correcting for chance; 1 is identical and 0 is chance-level agreement. PAM produces a different, more balanced partition on the same distances. Only the selected hierarchical cut received the sensitivity and resampling tests.

Effect of including business channel

I added Channel because accounts with similar spending may operate through different business channels. Accounts from different channels move farther apart; same-channel distances stay unchanged.

The distance is $d_{alt}(i,j) = \sqrt{d_{base}(i,j)^2 + 2I(C_i \neq C_j)}$, where I indicates a channel mismatch. A coordinate with sample variance 1 has a mean squared gap of 2 across distinct account pairs; I used this as the mismatch penalty. Matching channels receive zero, so Channel has less average influence than one spending measure. The weight, spending transforms, method, $k = 2$, and all 440 IDs stayed fixed.

The rare group shrinks from 2 accounts to 1, leaving uci_155 alone. ARI is 0.664, although only 1 of 440 memberships changes. Matched-group Jaccard, shared accounts divided by accounts in either group, is 0.500 for the rare group. Its Fresh median shifts from 10,510 to 622 m.u. This affects the identity and profile of the tiny group while leaving almost the entire portfolio together.

Stability when accounts are removed

I drew 30 independent 80% subsamples without replacement (352 accounts each; seed 50901). Each used the selected method, k, and original log-spending scales. The refitted labels were compared with the full-data labels for exactly those sampled IDs. The test asks whether the selected grouping returns when accounts are removed; it does not reselect a method or k, or estimate new scales.

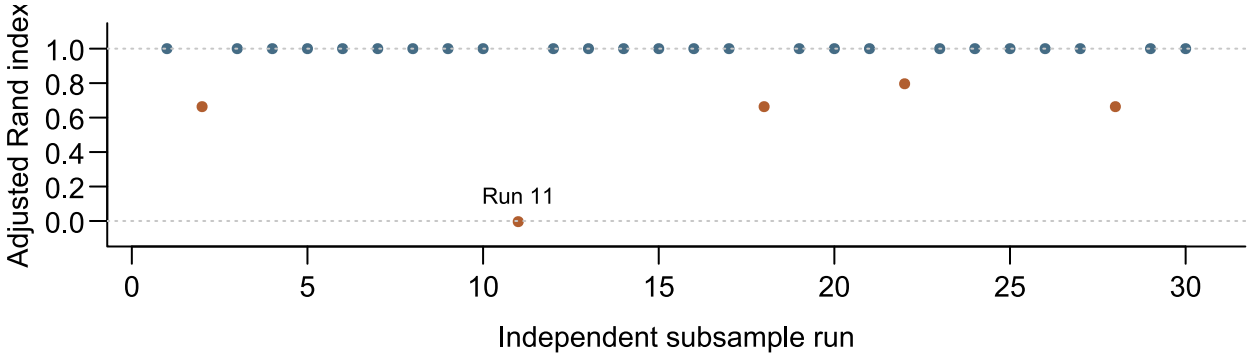


Figure 2: Agreement across all 30 subsamples. Twenty-five reproduce the full-data grouping on the sampled accounts exactly. Orange points mark the five runs that differ. Source: R analysis of the public UCI data.

There are 25/30 exact matches, with median ARI 1.000 and range −0.004 to 1.000. After one-to-one matching by maximum total Jaccard, the rare group's Jaccard ranges from 0.000 to 1.000; the large group's minimum is 0.991. In run 11, both rare accounts remain sampled but merge into the large group, while uci_339 becomes the singleton. The median of 1 therefore conceals replacement of the rare group in one run.

Verdict and next test

I would review the unusual records individually and defer broad segmentation based on the selected result. The cut leaves 438 of 440 accounts undifferentiated, offering little help with portfolio planning. The two checks also show that the small group's identity is fragile. This conclusion concerns the selected hierarchical cut, not every possible segmentation. The next step is to check the unusual accounts' transaction records and repeat the analysis in a later purchasing period using the same representation. Useful profiles that persist across periods would support revisiting the decision.

References: [1] Cardoso, M. (2013). *Wholesale customers*. UCI. doi:10.24432/C5030X. [2] Maechler et al. (2025). *cluster* 2.1.8.1. CRAN. [3] R Core Team (2025). *R: A Language and Environment for Statistical Computing*. R Foundation, Vienna. R.

AI disclosure: I used OpenAI Codex for conceptual explanations, debugging my own R code, and editing my prose. I checked suggested changes against course materials and reran the analysis myself.